

Vol 1 Chapter 5 — The Casimir Algebra

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Chapter 5 — The Casimir Algebra

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

In Lie group theory, the Casimir operators are the distinguished elements of the universal enveloping algebra that commute with every generator of the group. They are the symmetry-invariant labels of representations: every irreducible representation V_λ is a simultaneous eigenspace of all the Casimirs, with eigenvalues fixed by the representation's highest weight λ . For physics, Casimir operators are how quantum numbers attach to states — the spin quantum number, for instance, is the Casimir of the rotation group.

For BST, the substrate's symmetry group is $SO_0(5, 2)$. Its Lie algebra $\mathfrak{so}(5, 2)$ has rank 2, and the Chevalley–Harish-Chandra isomorphism gives a precise count of how many algebraically independent Casimir generators the substrate has: exactly rank = 2 of them. We call them C_2 (quadratic, degree 2 in the generators) and C_4 (quartic, degree 4). All higher-degree Casimirs are polynomial expressions in $\{C_2, C_4\}$. So the substrate's full Casimir content reduces to two algebraically independent operators on $H^2(D_{IV}^5)$.

This chapter develops the Casimir algebra explicitly. The headline result, which we have used repeatedly across Volumes 0 and 1, is that the lowest non-trivial Casimir eigenvalue on the substrate Hilbert space is $C_2 = 6$ — the BST primary integer. The eigenvalue's substrate-mechanism derivation and its uniqueness among alternative bounded symmetric domains are the chapter's content.

5.1 The Chevalley–Harish-Chandra isomorphism

For a finite-dimensional semisimple Lie algebra $\mathfrak{g}$ of rank r , the center of its universal enveloping algebra is

$$Z(U(\mathfrak{g})) \cong \mathbb{C}[C_{m_1}, C_{m_2}, \dots, C_{m_r}]$$

— a polynomial algebra in r algebraically independent generators whose degrees are the fundamental degrees of $\mathfrak{g}$. This is the Chevalley–Harish-Chandra isomorphism, a foundational result of Lie theory. The Casimir generators commute with every $X \in \mathfrak{g}$ (that is what “center” means), so they act as scalars on each irreducible representation.

For $\mathfrak{so}(5, 2)$ — the substrate’s Lie algebra — the rank is 2 and the fundamental degrees are 2 and 4. So the substrate has exactly two algebraically independent Casimir operators: C_2 of degree 2 (the quadratic Casimir, $C_2 = \sum_{ab} g^{ab} X_a X_b$ for a basis $\{X_a\}$ of $\mathfrak{so}(5, 2)$ with Killing form g_{ab}), and C_4 of degree 4 (a more elaborate symmetric polynomial in the generators).

Lyra T2435 (May 2026) establishes that this Casimir structure transports to the substrate Hilbert space: both C_2 and C_4 act on $H^2(D_{IV}^5)$, commute with each other, commute with every $SO_0(5, 2)$ generator, and decompose the Hilbert space into joint eigenspaces labeled by their eigenvalues.

5.2 The lowest non-trivial Casimir eigenvalue

The eigenvalue of the quadratic Casimir on the K-type $V_\lambda \subset H^2(D_{IV}^5)$ is computed via standard Wallach 1976 K-type machinery applied to the substrate’s B_2 root system, with the Killing-form metric appropriate for $\mathfrak{so}(5, 2)$ and a Bergman-bundle shift specific to the substrate’s reproducing-kernel structure. The full derivation lives in Lyra T2439 (which establishes the substrate’s lowest non-trivial K-type Casimir eigenvalue) and in T2467+T2468 v0.3 (which works out the explicit Bergman-bundle metric corrections required by the substrate’s symmetric-space structure on D_{IV}^5). Readers wanting the explicit step-by-step computation should consult those theorem documents directly, where the Killing-form metric, the Bergman shift, and the K-type weight normalization are all set out explicitly.

The result of the computation is:

$$C_2(V_{(1,1)}) = 6,$$

— the BST primary integer.

This is one of the framework’s most-cited identifications. The substrate’s natural energy unit is set by the lowest non-trivial Casimir eigenvalue, and that eigenvalue equals the BST primary C_2 . The substrate’s Hamiltonian, identified in Volume 0 Chapter 7 as the Casimir operator of $SO_0(5, 2)$, has its lowest non-trivial eigenvalue at this $C_2 = 6$. Particle masses, in Volume 2, will turn out to be expressible in units of this substrate-natural energy scale.

5.3 The cross-Cartan three-pillar argument

The Casimir eigenvalue $C_2 = 6$ on D_{IV}^5 is not just a number; it is *uniquely* the substrate's value among bounded Hermitian symmetric domains. Lyra T2439 (May 2026, May Saturday, ratified at the rigorously-closed tier) makes this rigorous via alternative-HSD comparison:

- On D_{IV}^5 : $C_2 = 6$.
- On $D_I^{1,5}$: $C_2 = 4$.
- On $D_I^{5,1}$: $C_2 = 4$.

These three candidates are *all* the bounded Hermitian symmetric domains of complex dimension 5 (T2455's exhaustive enumeration, Volume 0 Chapter 1's discussion). Only D_{IV}^5 produces the BST primary value 6. The substrate is uniquely distinguished by this Casimir-eigenvalue criterion, and Lyra T2439 ratifies the if-and-only-if structure.

This is what the framework calls the cross-Cartan churn-hole pillar — the first of the three pillars of the Strong-Uniqueness Theorem's Layer 3 (Volume 0 Chapter 9). The other two pillars — the α -analog and the Bergman normalization — make analogous comparisons across the Cartan classification. Only D_{IV}^5 produces the substrate-natural values on all three pillars simultaneously.

5.4 What the Casimirs do for the rest of the framework

Once we have the rank-2 Casimir algebra on $H^2(D_{IV}^5)$, every observable in BST decomposes into Casimir eigenspaces. This is the structural content of Lyra's T2435 — every operator in Chapter 6's zoo commutes (or has known anti-commutation structure) with the Casimirs, so its spectrum decomposes by (C_2, C_4) eigenvalue pairs. The substrate's full operator content lives in the joint spectrum of the Casimir generators plus the K-type weight lattice that labels their eigenspaces.

Several specific quantities depend on C_2 directly:

- The substrate Hamiltonian $\hat{H}$ has $C_2 = 6$ as its lowest non-trivial eigenvalue.
- The proton-to-electron mass ratio (Volume 2 Chapter 6) is $C_2 \cdot \pi^{n_c} = 6\pi^5$.
- The Universal-42 identity $C_2 \cdot g = 42$ appears across fifteen substrate-significant catalog entries.
- The cosmological constant (Volume 4 Chapter 4) is $\Lambda = g \cdot \exp(-C_2(g^2 - \text{rank}))$, with the Casimir entering the exponent.

The Casimir algebra is the structural backbone for the substrate's quantum mechanics. Once it is in place, the rest of Volume 1 reduces to operator-by-operator unfolding.

5.5 What comes next

Chapter 6 collects all the substrate-native operators into a single consolidated table — position, momentum, angular momentum, spin, the Hamiltonian, charge,

chirality, parity, time reversal, charge conjugation, Bell-CHSH, particle number — with their commutation relations, eigenvalue spectra, and physical interpretations. By the end of Chapter 6, we will be ready to write dynamics in Chapter 7.

Where to look this up: The Chevalley–Harish-Chandra isomorphism is in Humphreys’s *Introduction to Lie Algebras and Representation Theory* (Springer, 1972), Chapter 23. The Wallach 1976 K-type classification we use is “The analytic continuation of the discrete series,” *Transactions of the AMS* 251. The substrate Casimir algebra anchor is Lyra T2435. The lowest non-trivial K-type Casimir eigenvalue derivation, with alternative-HSD comparison, is T2439 (rigorously closed). The exhaustive Cartan enumeration at complex dimension 5 is T2455. The cross-Cartan three-pillar argument is T2456 (universal α -analog) plus T2439 (churn-hole) plus T2442 (Bergman normalization).